

LESSON 7-3 • HOMEWORK PACKET • SHEET 1 OF 2

Absolute Value as Distance

Maryland Standard 6.NS.C.7c • Learning target – I can use absolute value to describe how far a number is from zero.

REMEMBER

- **Absolute value** is a number's distance from 0. We write $|-7| = 7$.
- A number and its opposite have the same absolute value: $|5| = |-5| = 5$.
- Absolute value is never negative. A greater absolute value means **farther from 0**.

WORKED EXAMPLEWhich is farther from 0: -9 or 6 ?**Step 1** $|-9| = 9$ and $|6| = 6$.**Step 2** $9 > 6$, so -9 is farther from 0.**Answer** -9 **PRACTICE***show your work • label units*

1. Find each absolute value.

a. $|-14| = \underline{\hspace{2cm}}$

b. $|8| = \underline{\hspace{2cm}}$

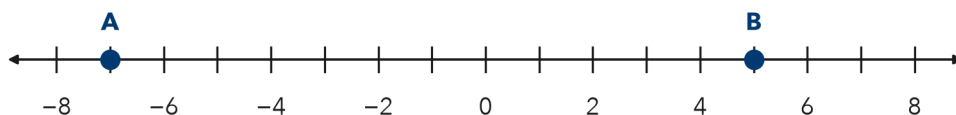
c. $|0| = \underline{\hspace{2cm}}$

d. $|-2.5| = \underline{\hspace{2cm}}$

e. $|3/4| = \underline{\hspace{2cm}}$

f. $|-1 \frac{1}{2}| = \underline{\hspace{2cm}}$

2. Use the number line. Find the absolute value of each point.



a. $|A| = \underline{\hspace{2cm}}$

b. $|B| = \underline{\hspace{2cm}}$

c. Farther from 0: $\underline{\hspace{2cm}}$

3. Compare. Write
- $<$
- ,
- $>$
- , or
- $=$
- in each blank.

a. $|-9| \underline{\hspace{1cm}} |7|$

b. $|-4| \underline{\hspace{1cm}} |4|$

c. $-10 \underline{\hspace{1cm}} |-3|$

d. $|-6.5| \underline{\hspace{1cm}} |6.5|$

e. $|-12| \underline{\hspace{1cm}} |-15|$

f. $0 \underline{\hspace{1cm}} |-1|$

- 4.
- Bank Balances.**
- Rosa's account balance is
- $-\$35$
- (she owes
- $\$35$
-). Ben's balance is
- $\$20$
- .

a. Find the absolute value of each balance. $\underline{\hspace{2cm}}$ b. Whose balance is farther from $\$0$? $\underline{\hspace{2cm}}$

- 5.
- Stretch.**
- List all the integers with an absolute value less than 3.

Integers $\underline{\hspace{10cm}}$

- 6.
- Reasoning.**
- Name two different numbers that have an absolute value of 4.5. Explain how you know.

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Ask someone at home: is it colder at -10 degrees or at 5 degrees? Which temperature is farther from 0?

LESSON 7-3 • HOMEWORK PACKET • SHEET 2 OF 2**Absolute Value as Total Distance**

Maryland Standard 6.NS.C.7c • Learning target – I can use absolute value to find the total distance between a negative and a positive reading.

REMEMBER

- **Opposite sides** of 0: **add** the absolute values.
- -3 to 5 is $3 + 5 = 8$ units.
- **Same side** of 0: **subtract** the absolute values.
- -9 to -2 is $9 - 2 = 7$ units.

WORKED EXAMPLE

A diver swims up from -24 feet to a boat deck at 6 feet. How far does she go?

Step 1 The numbers are on opposite sides of 0, so add.

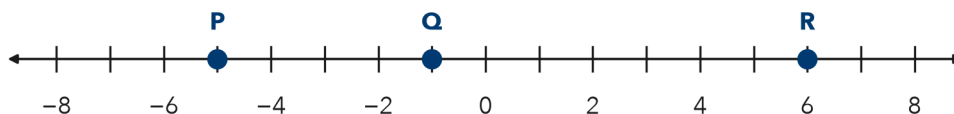
Step 2 $|-24| + |6| = 24 + 6 = 30$

Answer 30 feet

PRACTICE

show your work • label units

1. Use the number line. Find each distance.



a. P to R: _____

b. Q to R: _____

c. P to Q: _____

2. Find the distance between each pair of numbers.

a. -7 and 3 _____

b. -12 and -4 _____

c. -2.5 and 4 _____

3. **Temperature Swing.** It was -8 degrees at 6 a.m. and 15 degrees at 3 p.m. How much did it rise?

Answer _____ degrees

4. **Elevator.** An elevator goes from parking level -3 up to floor 9. Street level is floor 0. How many floors does it travel?

Answer _____ floors

5. **Find the Error.** To find the distance from -9 to 5 , Omar wrote $9 - 5 = 4$. Explain his mistake. Find the correct distance.

TEST PRACTICE

choose the best answer

6. What is the distance between -13 and 6 on a number line?

A 7

B 19

C -19

D -7

7. A cave floor is at -60 feet. A ledge is at -22 feet. How far apart are they?

A 82 feet

B 38 feet

C -38 feet

D 22 feet

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME With someone at home, find how many degrees the temperature changed today.